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# Artificial Intelligence, Cancer Nursing, and Sexual Health: Reassembling Care Networks in Digital Care

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## ABSTRACT

**Aims:** To examine how artificial intelligence (AI) reconfigures the conditions under which sexual health is recognized, articulated, and addressed in cancer nursing, thereby providing new conceptual insights into relational care, responsibility, and the organization of care networks.

**Design:** Discursive paper.

**Data Sources:** The paper draws on Cyber Nursing, the Fundamentals of Care framework, and actor–network theory, informed by empirical insights from a recent auto-netnographic study.

**Methods:** A theoretically informed discursive approach was used to develop conceptual analysis and synthesis across nursing theory and sociomaterial perspectives.

**Results:** AI is conceptualized as a nonhuman actor that participates in care networks by facilitating the articulation and visibility of sensitive concerns such as sexual health and by enabling care-like enactments. The analysis introduces the concepts of weakly networked care and reassembling care networks to explain how AI reshapes the sociomaterial conditions through which fundamental care is recognized, articulated, and integrated into nursing practice.

**Conclusion:** AI can make sexual health more visible and discussable in cancer nursing, but it does not constitute ethical care, as responsibility, relational continuity, and integration remain grounded in nursing practice. Rather than challenging the Fundamentals of Care framework, the concept of reassembling care networks complements its continued development by directing attention towards the changing sociomaterial conditions through which fundamental care is enacted.

## 1 | Introduction

Artificial intelligence (AI) is increasingly embedded in oncology care, where it is often framed in the language of efficiency, prediction, and technical optimization (O'Connor et al. 2024). While these developments are significant, they risk obscuring a more fundamental transformation: AI is also reshaping how care is enacted, experienced, and recognized. This becomes particularly visible in domains that are relational, sensitive, and historically under-addressed—such as sexual health.

Sexual health is central to life during and after cancer (Laursen et al. 2026; Madsen et al. 2025; Muntlin Athlin et al. 2018; Wettergren et al. 2017). Treatment may affect function, intimacy, identity, and relationships, often in ways that persist beyond clinical recovery. Despite this, sexual health remains inconsistently addressed in cancer nursing practice (Papadopoulou et al. 2019). The omission of sexual health is theoretically consequential because it exposes a gap between nursing's commitment to holistic care and what is routinely enacted.

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This paper examines how AI reconfigures the conditions under which sexual health is recognized and addressed in cancer nursing. While existing cancer nursing literature has primarily examined the application and implementation of AI technologies (O'Connor et al. 2024), less attention has been directed towards how AI reshapes the articulation and integration of fundamental care needs, such as sexual health. Drawing on Cyber Nursing, the Fundamentals of Care (FoC), and actor–network theory (ANT) and informed by Christiansen et al. (2026), the paper develops a sociomaterial perspective on how AI participates in care networks. The empirical study serves as an illustrative case informing theoretical development rather than as a source of primary empirical analysis. Through this analysis, the paper introduces three interconnected concepts—care-like enactments, weakly networked care, and reassembling care networks—that provide the analytical basis for examining how AI reshapes the pathways through which sexual health enters nursing care.

The aim of this discursive paper is to examine how artificial intelligence reconfigures the conditions under which sexual health is recognized, articulated, and addressed in cancer nursing, thereby providing new conceptual insights into relational care, responsibility, and the organization of care networks.

### 1.1 | Approach to the Discursive Analysis

This paper adopts a discursive approach that is analytical and theory-building rather than empirical. Its purpose is to develop conceptual understanding of how artificial intelligence reconfigures the conditions under which sexual health is recognized, articulated, and addressed in cancer nursing through critical engagement with nursing theory, sociomaterial perspectives, and an emerging area of practice.

The paper was developed through an iterative process of theoretical comparison and conceptual refinement focused on a specific nursing problem: the persistent marginalization of sexual health in cancer care, despite its recognition as an important aspect of holistic and person-centred care (Feo et al. 2018; Laursen et al. 2026).

To examine this problem, three complementary theoretical perspectives were selected for their ability to illuminate different dimensions of care. Cyber Nursing (Eriksson and Salzmann-Erikson 2013, 2016) was included for its framework for understanding how care is enacted in digitally mediated environments. The Fundamentals of Care framework (Feo et al. 2018; Kitson 2018) was selected for its nursing-specific understanding of care as the integration of physical, psychosocial, and relational needs. Actor–network theory (Latour 2005; Mol 2008; Suchman 2007) was incorporated for its capacity to analyse how care practices are produced and sustained through relations among human and nonhuman actors.

These perspectives were not combined to produce a unified theory. Rather, they were brought into dialogue to generate new insights into how AI participates in contemporary care practices. Together, they provide insight into mediated care, the content and integration of care, and the sociomaterial conditions through which care practices become stabilized, marginalized, or reconfigured.

The paper is informed by empirical insights from Christiansen et al. (2026), an auto-netnographic study of interactions with generative AI chatbots for sexual health information after prostate cancer among men who have sex with men. The study was not re-analysed as a source of primary data. Instead, it served as an illustrative empirical case through which the theoretical implications of AI-mediated interactions could be explored and elaborated.

Through this process, the paper develops the concepts of care-like enactments, weakly networked care, and reassembling care networks as theoretical contributions intended to support future nursing scholarship, the continued development of the Fundamentals of Care framework, and research concerning AI-mediated care.

## 2 | Theoretical Foundations

### 2.1 | Cyber Nursing: Mediated Presence and Changing Conditions of Care

Cyber Nursing (Eriksson and Salzmann-Erikson 2013, 2016) provides a conceptual starting point for understanding how nursing practice is extended into digital environments. Central to this framework is a shift away from viewing care as dependent on physical co-presence towards understanding care as enacted through communication, responsiveness, and relational engagement across mediated contexts.

This repositioning is not merely a technical adjustment but a conceptual one. It challenges the assumption that presence is inherently tied to physical proximity, suggesting instead that presence is constituted through interaction. Within this perspective, digital environments are not external to care but are sites where care is performed, negotiated, and recognized.

The relevance of Cyber Nursing becomes more pronounced with the emergence of generative AI. While earlier digital technologies primarily enabled communication among human actors, AI introduces a form of interaction in which responses are generated by nonhuman systems. This raises questions about how presence is enacted when responsiveness is partially delegated to algorithmic processes. If presence is understood as communicative responsiveness rather than physical proximity, AI complicates how such responsiveness is interpreted and evaluated within nursing practice.

Rather than resolving these questions, Cyber Nursing provides a framework for recognizing that the conditions of care are shifting. It invites consideration of how mediated interaction reshapes not only where care occurs, but how it is experienced and attributed. In the present paper, Cyber Nursing serves as the conceptual bridge between relational nursing theory and AI-mediated care by explaining how the conditions under which care is initiated and experienced may change in digitally mediated environments without assuming physical co-presence.



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## 2.2 | Fundamentals of Care: The Content and Integration of Care

The Fundamentals of Care (FoC) framework (Feo et al. 2018; Kitson et al. 2010) provides a complementary perspective by specifying what care should encompass. It conceptualizes care as the integration of physical, psychosocial, and relational dimensions within a trusting nurse–patient relationship. Care is not defined by isolated tasks, but by how these dimensions are brought together in practice (Kitson 2018).

Sexual health traverses all three domains. It involves bodily changes and function, emotional and identity-related experiences, and relational aspects such as intimacy and partnership. Laursen et al. (2026) argue that sexuality should therefore be understood as a fundamental care need, rather than an optional or specialized concern.

However, the integration emphasized in FoC also highlights a tension between conceptual ideals and clinical realities. While sexual health can be positioned as fundamental, its enactment remains inconsistent. This inconsistency becomes visible when considering how fundamental care is defined and operationalized over time.

Muntlin Athlin et al. (2018) explicitly identify sexuality as part of fundamental care needs within cancer care. Yet, in subsequent efforts to standardize the concept (Feo et al. 2018), this specificity becomes less pronounced. Sexuality is not necessarily excluded, but it becomes less visible within broader formulations. This suggests that even within frameworks committed to holistic care, certain needs may lose clarity and emphasis as concepts are stabilized.

From this perspective, FoC not only defines what care should include but also provides a framework for examining how fundamental care is—or is not—realized in practice. Recent discussions have emphasized the importance of moving beyond defining fundamental care towards understanding the conditions that enable its consistent enactment (Kitson 2026). Sexual health therefore becomes a useful lens through which to examine how integration is achieved—or fails to be achieved—in everyday nursing practice.

## 2.3 | Actor–Network Theory: Care as Relational Achievement

Actor–network theory (Latour 2005; Mol 2008; Suchman 2007) offers a way of analysing how care is produced through relations between human and nonhuman actors. Rather than treating care as something that is delivered by individuals, ANT conceptualizes care as an effect of interactions within networks that include people, technologies, routines, and institutional structures.

Within this perspective, AI can be understood as a nonhuman actor that participates in shaping care processes by influencing how concerns are articulated, how interactions unfold, and what becomes possible to express within a given encounter. Importantly, participation does not imply agency in a moral sense. AI does not take responsibility, but it does shape action.

A key insight from ANT is that practices become stable when they are supported by networks that align actors, expectations, and material arrangements. Conversely, practices remain fragile when such alignment is weak or inconsistent. This provides a way of understanding why certain aspects of care—such as medication management—are highly routinised, while others—such as sexual health—remain variable and contingent.

ANT therefore shifts the analytical focus from individual competencies and attitudes towards the sociomaterial arrangements through which care becomes possible. It allows questions of marginalization to be reframed as questions of enrollment, stability, and relational alignment.

## 2.4 | Theoretical Integration: Towards a Sociomaterial Understanding of Care

Taken together, Cyber Nursing, the Fundamentals of Care, and actor–network theory offer complementary but distinct analytical lenses. Cyber Nursing foregrounds the conditions under which care is mediated; FoC specifies the content and integration of care; and ANT explains how care is produced and stabilized through networks.

These three perspectives address complementary questions. Cyber Nursing asks under what conditions care may occur in digitally mediated environments. The Fundamentals of Care framework asks what constitutes fundamental care, while actor–network theory examines how such care is produced and sustained through evolving sociomaterial networks.

Their integration allows for a more comprehensive understanding of how care is both conceptualized and enacted. Cyber Nursing highlights that care increasingly occurs in digitally mediated contexts. FoC insists that such care must remain holistic and relational. ANT provides the tools to examine how these aspirations are realized—or undermined—within concrete sociomaterial arrangements.

These perspectives also resonate with broader developments in nursing scholarship concerning technology and care. Cyber Nursing has highlighted how caring practices increasingly extend into digitally mediated environments and how presence, communication, and responsiveness are reconfigured through technology (Eriksson and Salzmann-Erikson 2016). More recently, AI has emerged as an important topic within cancer nursing, with research largely focusing on the implementation and application of AI technologies to support clinical decision-making and care delivery (O'Connor et al. 2024). However, less attention has been directed towards understanding AI as part of the sociomaterial arrangements through which care is enacted. By bringing together Cyber Nursing, the Fundamentals of Care, and actor–network theory, the present paper contributes to this emerging conversation by examining how generative AI participates in shaping the visibility, articulation, and integration of sexual health within cancer nursing.

This integrated framework makes it possible to conceptualize AI not simply as a tool that supports care, but as an actor that participates in shaping care networks. It also enables analysis



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of how certain needs, such as sexual health, may be theoretically recognized yet remain weakly integrated in practice.

Building on this integrated perspective, the following analysis examines how AI participates in the articulation of sexual health concerns, how these interactions may be understood as forms of weakly networked care, and how they contribute to the reassembling of care networks within cancer nursing.

## 3 | Sexual Health in Cancer Care: A Marginalized and Weakly Networked Fundamental Need

Sexual health is widely recognized as an integral component of well-being in cancer care, encompassing physical function, identity, intimacy, and relational life. Cancer treatments frequently affect these dimensions in ways that extend beyond the immediate clinical trajectory, shaping how individuals experience themselves and their relationships over time. Within the Fundamentals of Care framework, sexual health reflects the integration of physical, psychosocial, and relational dimensions of care rather than separate domains (Feo et al. 2018; Kitson et al. 2010). Laursen et al. (2026) explicitly articulate sexuality as a fundamental care need, reinforcing its conceptual centrality within the framework.

Despite this theoretical positioning, sexual health remains inconsistently addressed in clinical practice and is frequently reported as insufficiently integrated into routine cancer care (Papadopoulou et al. 2019; Reese et al. 2017). Previous research has identified several barriers to addressing sexual health, including limited knowledge, uncertainty about professional responsibility, lack of confidence, time constraints, and discomfort initiating conversations about sexuality (Papadopoulou et al. 2019). Although these factors are important, they provide only a partial explanation for why sexual health continues to occupy a marginal position within cancer nursing despite broad professional agreement that it constitutes a fundamental aspect of holistic care. This gap between theoretical recognition and everyday enactment suggests that sexual health is not absent from nursing knowledge but rather insufficiently stabilized within the networks that organize and sustain routine care.

Interventions such as Fex-Talk (Winterling et al. 2020) demonstrate that even relatively brief educational efforts can enhance nurses' readiness to engage in conversations about sexuality. Similarly, structured approaches such as the BETTER model (Åkerstrom Wenneberg et al. 2026) provide frameworks intended to normalize and support communication about sexual health. These interventions are important because they demonstrate that competence can be developed and communication can be facilitated.

However, what these studies also reveal—perhaps more implicitly than explicitly—is that increased readiness or the availability of structured models does not in itself stabilize sexual health as a routine component of care. Engagement with sexuality remains variable, contingent, and often dependent on individual initiative. This suggests that the marginalization of sexual health is not solely a matter of individual capability but reflects how care practices are organized and prioritized within broader clinical contexts.

From an actor–network perspective, this can be understood as a question of how sexual health is enrolled within care networks. Practices that become routine are typically supported by a constellation of elements: documentation systems, institutional expectations, professional roles, time allocation, and shared communicative norms. In contrast, sexual health often lacks this level of enrollment. It is recognized in principle but is insufficiently embedded in the structures that make care durable in everyday practice.

This condition can be described as weakly networked care: a situation in which a care need is theoretically recognized but insufficiently embedded within the organizational, relational, and material structures required for its consistent enactment. Sexual health is not absent from nursing knowledge, nor is it entirely excluded from practice. Rather, it occupies an unstable position in which its relevance is acknowledged, but its enactment is inconsistent. It may emerge in certain encounters, depend on particular practitioners, or be triggered by specific cues, but it does not consistently form part of the expected and supported repertoire of care.

This perspective also sheds light on developments within nursing theory itself. While Muntlin Athlin et al. (2018) explicitly identified sexuality as part of fundamental care needs in cancer care, subsequent efforts to standardize the concept (Feo et al. 2018) do not maintain this level of specificity. Sexuality becomes less visible within broader formulations of fundamental care, even as the commitment to holistic care remains. This suggests that marginalization can occur not only in practice but also within processes of conceptual consolidation.

What emerges, therefore, is not a simple gap between theory and practice, but a more complex dynamic in which certain needs are more or less strongly stabilized within care networks. Sexual health illustrates how a need can be theoretically recognized yet remain practically fragile. Its marginal position is sustained through the absence of consistent structures that support its articulation, documentation, and follow-up.

Understanding sexual health as weakly networked care shifts the focus of analysis. Rather than asking why nurses do not address sexuality, the question becomes how care networks can be configured so that sexual health is more consistently recognized and enacted. This perspective provides the analytical foundation for examining how AI may participate in reshaping these networks—not by replacing nursing care, but by altering the conditions under which sexual health becomes visible, articulated, and integrated into practice.

## 4 | AI As a Nonhuman Actor in Sexual Health Care

The study by Christiansen et al. (2026) offers an empirically grounded point of departure for examining how AI participates in sexual health care. Rather than approaching AI as a



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tool for information delivery, the auto-netnographic analysis illustrated how interactions with generative AI chatbots extended beyond information provision and assumed relational qualities. Across the AI-mediated interactions analysed, the chatbots frequently generated responses that users could experience as responsive, attentive to concerns regarding erectile dysfunction, intimacy, and sexual identity, and validating users' experiences by acknowledging these concerns as legitimate. This suggests that AI-mediated communication may function relationally even in the absence of a human interlocutor.

From an actor–network perspective, this observation shifts how AI is conceptualized. Rather than functioning as an external support to care, AI can be understood as a nonhuman actor that participates in shaping the interaction itself. It contributes to how questions are formulated, how concerns are recognized, and what kinds of responses become available. In this sense, AI does not simply respond to pre-existing needs; it is involved in configuring how those needs are articulated in the first place.

This becomes particularly relevant in the context of sexual health, where articulation is often constrained. Concerns related to sexuality may remain unspoken in clinical encounters due to embarrassment, anticipated judgement, or uncertainty about relevance. This may include concerns about erectile dysfunction, altered intimacy, changes in sexual identity, or uncertainty about sexual relationships following treatment. The study suggests that AI-mediated interaction can alter these conditions by providing a space in which such concerns can be expressed more readily. The perceived non-judgmental character of AI-mediated interaction appears to lower thresholds for disclosure.

To capture the character of these interactions, the notion of care-like enactments is useful. These enactments emerge when interactions produce effects commonly associated with care—such as validation, acknowledgment, or perceived empathy—but are not grounded in professional responsibility or relational continuity. These qualities do not reside within the AI system itself but emerge through the interaction between user, system, and context, shaped by expectations, prior experiences, and the framing of the exchange.

The distinction between care and care-like enactments is analytically important. If AI-mediated interactions are understood as equivalent to care, there is a risk that the ethical and relational foundations of nursing practice become obscured. At the same time, reducing these interactions to mere information exchange fails to account for their experiential significance. What emerges instead is a form of interaction that occupies an intermediate position: meaningful for users, yet insufficient as care in a professional and ethical sense. For example, a patient may experience an AI-generated response as validating and reassuring, yet no healthcare professional assumes responsibility for interpreting, documenting, or following up the concern.

At the same time, the capacity of AI to facilitate articulation should not be conflated with the provision of care. While AI may enable the initial expression of concerns, it does not ensure that these concerns are followed up, contextualized, or integrated into ongoing care processes. The interaction is episodic and lacks continuity, and it does not carry responsibility for outcomes. From this perspective, AI can be seen as participating in the early stages of care—when concerns become visible—but unable to sustain care over time.

Furthermore, AI-mediated interactions are shaped by the sociotechnical conditions under which they are produced. The inclusivity and responsiveness illustrated in the auto-netnographic analysis depend on how systems are trained, the data they rely on, and the assumptions embedded in their design. Consequently, AI does not simply respond to concerns but may also reproduce or challenge prevailing norms regarding sexuality, relationships, and whose experiences become recognized. As such, AI's participation in care networks is neither neutral nor universally beneficial, but contingent upon the sociotechnical configurations through which knowledge is generated and communicated.

Taken together, these observations suggest that AI does not merely support existing care practices but participates in reshaping the conditions under which sexual health becomes visible and discussable. It contributes to a reconfiguration of care networks by enabling articulation where silence has prevailed. However, this participation remains partial. AI may open the conversation by enabling concerns to be articulated before they enter clinical care, but it does not complete it. The work of transforming articulated concerns into accountable, relational, and continuous care remains the responsibility of nursing practice.

## 5 | Reassembling Care Networks: Theoretical Tensions

The integration of AI into cancer nursing does not simply add a new tool to existing practices; it introduces a set of tensions that challenge how care is conceptualized, enacted, and evaluated. These tensions are not incidental but arise from the ways in which AI reconfigures relationships between actors, alters communicative conditions, and redistributes aspects of interaction within care processes.

One central theoretical tension concerns the relationship between care and care-like enactments. Care-like enactments should be understood as interactional effects that resemble care without carrying its ethical obligations. Reassembling care networks refers to the broader reconfiguration of relations through which such interactions become possible and consequential. As illustrated in the preceding section, AI-mediated interactions may produce effects that resemble care—such as validation, responsiveness, and perceived empathy. However, these effects are not grounded in responsibility, continuity, or professional judgement. These AI-mediated interactions create a situation in which interactions may feel supportive without being anchored in accountable care. The tension is therefore not merely definitional but also practical: how should such interactions be interpreted within nursing practice, and what status should they be given in relation to care?



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A second tension emerges between accessibility and relational depth. AI may enable users to articulate concerns that would otherwise remain unspoken, particularly in areas such as sexual health, where barriers to communication are well documented. In this sense, AI can expand access to forms of interaction that approximate support. However, this expansion is accompanied by a potential thinning of relational depth. AI-mediated interactions are typically episodic, decontextualized, and not embedded in ongoing relationships. What is gained in accessibility may therefore be accompanied by a loss of continuity, contextual understanding, and sustained engagement.

This tension can be further understood as a shift in where and how care begins. AI may facilitate the initial articulation of concerns, effectively moving the point of entry into the care process. However, it does not ensure that these concerns are taken forward into structured, relationally grounded care. The result is potential fragmentation, in which articulation and follow-up are distributed across different parts of the care network without clear integration.

A third tension relates to visibility and integration. As discussed earlier, sexual health in cancer care can be understood as weakly networked, in that it is inconsistently integrated into clinical routines and communicative practices. AI has the potential to increase the visibility of such concerns by making them easier to articulate. However, increased visibility does not automatically translate into integration within care processes. Concerns may be expressed more frequently without being systematically addressed, documented, or followed up on.

From an actor–network perspective (Latour 2005; Mol 2008), this can be understood as a partial reconfiguration of the network. AI may strengthen certain connections—particularly those enabling the articulation of sexual health concerns—while leaving others underdeveloped, especially those supporting continuity, follow-up, and accountability. The network is thus reassembled but not necessarily stabilized.

A further tension concerns agency and responsibility. AI participates in shaping interactions yet does not bear responsibility for the outcomes of those interactions. This creates a redistribution of action without a corresponding redistribution of accountability. While AI may influence how concerns are expressed and responded to, responsibility for interpretation, follow-up, and ethical judgement remains with human actors. This asymmetry raises questions about how responsibility is recognized and managed within increasingly complex care networks.

These tensions can be brought together through the concept of reassembling care networks. Rather than viewing AI as either enhancing or undermining care, this concept captures how AI participates in reconfiguring the relations through which care is enacted. Reassembling care networks involves shifts in what becomes visible, how concerns are articulated, where interactions occur, and how responsibilities are distributed.

Importantly, reassembly should not be understood as a completed or stable transformation. It is an ongoing and uneven process in which certain aspects of care may be strengthened while others become attenuated. AI may enable new forms of engagement while simultaneously challenging established forms of relational care. The task for nursing is therefore not to resolve these tensions by choosing between technological and relational approaches, but to engage with how these configurations can be shaped in ways that support the recognition and integration of fundamental care needs.

In this sense, the introduction of AI does not resolve the marginalization of sexual health in cancer care. Rather, it makes visible the conditions under which such marginalization occurs while simultaneously creating new opportunities and new tensions. The analytical value of focusing on tensions is therefore not to identify problems to be eliminated, but to understand how care is continuously negotiated within evolving sociomaterial arrangements. These tensions suggest that AI does not simply enter care as an additional actor but reconfigures the very conditions under which care can be recognized, enacted, and evaluated.

The present analysis also contributes to ongoing discussions about the relevance of nursing theory in the context of artificial intelligence. Recent scholarship has emphasized the need to preserve core nursing values and relational commitments amid growing digital transformation (Nashwan and Kunjavara 2026). From this perspective, theories such as Watson's Theory of Human Caring and Peplau's Interpersonal Relations Theory remain highly relevant because they position caring relationships and human engagement at the centre of nursing practice. Rather than challenging these foundations, the present paper builds on them by addressing a complementary question: how care-related processes emerge in increasingly mediated environments where nonhuman actors participate in shaping communication, visibility, and access to care.

While Watson and Peplau focus primarily on human relationships, the current analysis examines how AI may influence the conditions under which such relationships are initiated, sustained, or disrupted. In this sense, AI cannot replace caring relationships, but it may participate in reconfiguring the networks through which caring relationships become possible. This perspective is consistent with emerging work applying established nursing theories to technologically mediated care. For instance, Andtfolk et al. (2026) draw on Peplau's Interpersonal Relations Theory to examine social robot-assisted medication counselling, illustrating the continued relevance of relational nursing theories in interactions involving nonhuman actors.

Taken together, these tensions describe the conceptual development through which AI reshapes the pathways by which sexual health enters nursing care. The conceptual development presented in the preceding analysis is summarized in Figure 1.

## 6 | Ethical Reflections: Responsibility, Relationality, and the Limits of AI-Mediated Care

In this paper, ethical care is understood as care characterized by responsibility, relational continuity, and integration within an ongoing care trajectory. This understanding is informed by Eriksson's caritative theory of caring, in which care is grounded in responsibility for the suffering person and in the affirmation



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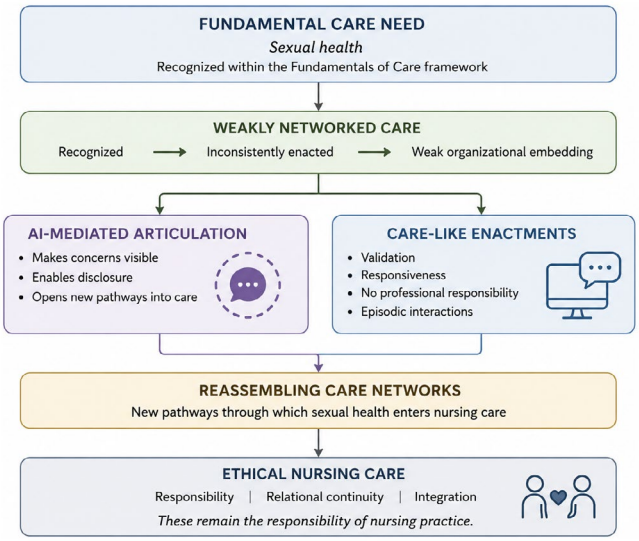



**FIGURE 1** | Conceptual development of AI-mediated reassembling of care networks in cancer nursing. Colours are used solely to distinguish successive conceptual components of the model and do not indicate separate theoretical categories.

of human dignity. Ethical care therefore extends beyond supportive interactions to encompass an enduring commitment to the person's well-being and care needs. Consistent with Mol's (2008) account of care as an ongoing relational practice rather than a discrete intervention, ethical care is understood as something that is sustained through continuous engagement, response, and integration within care processes.

A first ethical consideration concerns the relationship between action and responsibility. As discussed earlier, AI participates in shaping interactions by influencing how concerns are articulated and how responses are formulated. In this sense, action is distributed across human and nonhuman actors. However, this distribution of action is not matched by a distribution of responsibility. AI does not bear accountability for the outcomes of its responses, nor can it be held responsible for omissions, misunderstandings, or harm. Responsibility remains situated within human actors and institutional structures, even as the pathways through which care unfolds become increasingly mediated.

This asymmetry introduces a form of ethical opacity. When interactions are co-produced by human and nonhuman actors, it may become less clear how responsibility is to be attributed and enacted in practice. For nursing, this raises questions not only about accountability, but also about professional identity. If aspects of interaction are delegated to AI, what remains the distinct ethical role of the nurse? This is not a question of replacement, but of redefinition: how responsibility is maintained in contexts where action is distributed.

This distinction between participation and responsibility aligns with discussions about the use of technology as a care resource. Drawing on Eriksson's caring science tradition, Andtfolk (2022) argues that technologies may contribute to caring processes and support care encounters, while responsibility for care remains grounded in human relationships and professional practice. From this perspective, AI may participate in the conditions under which care becomes possible without itself becoming the bearer of caring responsibility.

A second ethical tension concerns what may be described as simulated empathy. Generative AI systems can produce responses that resemble empathic communication through tone, phrasing, and structure. Users may experience these responses as supportive, particularly in situations where they feel hesitant to disclose sensitive concerns to a human professional. However, such responsiveness is not grounded in lived experience, moral commitment, or relational continuity.

The ethical issue is therefore not that AI fails to provide empathy, but that it can simulate empathy in ways that may reorganize how support is perceived and trusted. If users experience AI interactions as empathic, this may influence expectations of care, potentially blurring distinctions between relational engagement and simulated responsiveness. From a nursing perspective, this raises questions about what constitutes authentic care and how trust is established and sustained. This also includes recognizing that AI-generated responses may be incomplete or misleading, requiring professional judgement in their interpretation. These tensions suggest that AI does not simply enter care as an additional actor but reconfigures the very conditions under which care can be recognized, enacted, and evaluated.

These questions also resonate with longstanding discussions about the relationship between technology and caring in nursing. As Sandelowski (1999) argues, technology should not be understood as inherently opposed to caring. However, the growing ability of AI systems to simulate empathetic interaction raises new questions about how caring is recognized, interpreted, and attributed in technologically mediated encounters.

A third consideration concerns relational continuity. Nursing care is defined not only by individual interactions but also by the ongoing development of relationships in which knowledge of the patient accumulates over time. AI-mediated interactions, by contrast, are typically episodic and context-limited. While they may enable the articulation of concerns, they do not carry forward responsibility for those concerns across encounters.

This lack of continuity has ethical implications. It creates a situation in which concerns may be expressed without being integrated into a broader care trajectory. From the perspective of the patient, this may result in a form of fragmented care in which support is momentarily available but not sustained. From a nursing perspective, it reinforces the importance of translating AI-mediated articulation into relationally grounded follow-up.

A fourth ethical dimension relates to inclusion and exclusion. AI has the potential to provide more inclusive responses, particularly in relation to sexual minority experiences that may be insufficiently addressed in clinical encounters. The autonetnographic analysis by Christiansen et al. (2026) illustrated how generative AI chatbots generated responses that engaged with sexual minority perspectives in ways that could be experienced as affirming. However, this inclusivity is contingent upon the data and assumptions embedded within AI systems.



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AI may therefore both challenge and reproduce existing inequalities. It can make certain experiences more visible while leaving others unrepresented or misrepresented. Ethical engagement with AI in nursing must therefore include attention to how different forms of knowledge and experience are encoded within these systems, and whose perspectives are amplified or marginalized.

Taken together, these considerations suggest that the ethical implications of AI in cancer nursing cannot be reduced to questions of technological performance or regulation alone. Instead, they point to the need for a relational and practice-oriented ethics in which AI is understood as part of a broader sociomaterial configuration of care.

From this perspective, the role of nursing is not diminished but becomes more complex. Nurses remain responsible for ensuring that care is relational, accountable, and responsive to individual needs. This includes interpreting AI-mediated interactions, contextualizing them within broader care processes, and addressing the limitations inherent in care-like enactments.

Ethical practice in the context of AI, therefore, involves active engagement rather than passive adoption. It requires ongoing reflection on how AI participates in care, how responsibility is maintained, and how relational care can be sustained within increasingly mediated environments. In this sense, ethics is not an external framework applied to technology, but an integral part of how care is continuously negotiated and enacted.

## 7 | Implications for Nursing: Fundamental Care Under Changing Conditions

The conceptual developments presented in this paper have implications that extend beyond the implementation of AI technologies in cancer nursing. Rather than asking whether AI should be incorporated into practice, the analysis shifts attention towards how fundamental care is recognized, initiated, and sustained under changing sociomaterial conditions. From this perspective, AI does not diminish the relevance of the Fundamentals of Care framework; rather, it highlights the need to continue developing the framework so that it can account for the increasingly mediated conditions through which nursing care is enacted.

### 7.1 | Clinical Practice: From Articulation to Integration

One of the most important implications of this analysis concerns where care begins. AI-mediated interactions may enable concerns—particularly those relating to sexual health—to be articulated before patients encounter a nurse. This does not diminish the importance of nursing; rather, it shifts attention towards how these articulations are recognized, interpreted, and integrated into ongoing care.

From this perspective, AI-mediated interactions should not be viewed as external to nursing practice but as part of the evolving pathways through which care needs emerge. Patients may increasingly arrive at consultations having already explored sensitive concerns using generative AI. These interactions can offer valuable opportunities for assessment and dialogue if nurses recognize them as clinically relevant rather than peripheral to nursing care. Questions such as "Have you used digital tools or AI systems to seek information about this concern?" can help identify needs that have not previously been discussed in clinical conversations.

The practical implication is therefore not that AI should replace conversations about sexual health, but that nursing should remain responsible for transforming articulated concerns into accountable, relational care. AI may contribute to making concerns visible and discussable, but it cannot ensure continuity, contextual understanding, or professional follow-up. These remain fundamental nursing responsibilities.

From the perspective of the Fundamentals of Care framework, this suggests that future nursing practice will increasingly involve integrating concerns that arise across multiple interactional contexts rather than solely within face-to-face encounters. The challenge is therefore not whether AI can care, but how nursing can sustain fundamental care under increasingly mediated conditions.

### 7.2 | Reasserting Sexual Health as Fundamental Care

Conceptual recognition alone does not ensure that a care need becomes part of everyday nursing practice. The analysis of sexual health as weakly networked care suggests that fundamental care requires not only conceptual clarity but also organizational, relational, and communicative structures that enable its consistent enactment.

From this perspective, strengthening sexual health within cancer nursing is not primarily a matter of improving individual competence. It also requires routines, documentation practices, communicative norms, and organizational expectations that make sexual health an anticipated and integrated component of care rather than an optional addition. The comparison between Muntlin Athlin et al. (2018) and Feo et al. (2018) illustrates how the visibility of sexuality may shift even within frameworks committed to holistic care, reinforcing the importance of continually making fundamental care needs explicit.

The implications extend beyond sexual health. If the Fundamentals of Care framework is to remain responsive amid increasingly mediated conditions of care, attention must focus not only on what constitutes fundamental care but also on the conditions through which it becomes embedded and sustained in practice. AI may help make concerns more visible, but without corresponding integration into nursing practice, such visibility remains transient rather than transformative.

### 7.3 | Nursing Education: Preparing Nurses for Mediated Care

The emergence of AI-mediated care has implications that extend beyond technical competence. Nursing education must prepare future nurses to understand how care is increasingly initiated,



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articulated, and negotiated across digitally mediated environments. This includes recognizing the distinction between care and care-like enactments, critically interpreting AI-mediated interactions, and understanding their implications for professional responsibility.

Competence in addressing sexual health remains equally important. Educational initiatives such as Fex-Talk (Winterling et al. 2020) demonstrate that nurses' readiness to discuss sexuality can be strengthened. However, nursing education must integrate this competence with critical AI literacy so that nurses are equipped not only to communicate about sensitive issues but also to critically interpret how AI participates in shaping those interactions.

From the perspective of the Fundamentals of Care framework, this suggests that nursing education should increasingly prepare students to understand care as both relational and sociomaterial. Future nurses will need to navigate environments in which communication, technology, and professional judgement become progressively intertwined.

### 7.4 | Research: Examining Care Under Changing Conditions

The analysis presented in this paper suggests that future research should move beyond evaluating AI as an isolated intervention and instead examine how AI participates in the reconfiguration of care networks. This includes investigating the articulation of concerns, how these concerns enter clinical care, and how responsibility is distributed across human and technological actors.

Such research also requires greater attention to how AI systems produce inclusion and exclusion, particularly regarding diverse sexual identities, experiences, and forms of embodiment. From an actor–network perspective, these questions concern not only individual competencies but also the organizational, relational, and material conditions through which care becomes possible.

More broadly, future nursing research may benefit from examining how fundamental care is sustained under increasingly mediated conditions. AI provides one illustration of these developments, but the underlying challenge concerns how nursing theory can continue to account for the changing sociomaterial conditions through which care is recognized, articulated, and enacted.

### 7.5 | Future Development of the Fundamentals of Care Framework

The present analysis suggests that the continued development of the Fundamentals of Care framework requires increasing attention to the sociomaterial conditions through which care becomes possible. While the Fundamentals of Care framework has provided nursing with a powerful language for describing holistic and relational care, AI reveals that fundamental care is also shaped by changing configurations of technologies, communication, and distributed forms of interaction. Rather than challenging the framework, the concept of reassembling care networks extends it by directing attention towards how fundamental care is initiated, recognized, and integrated within increasingly mediated environments.

## 8 | Conclusion: Reassembling Care Networks in AI-Mediated Cancer Nursing

This paper has examined how the integration of AI into cancer nursing reshapes the conditions under which sexual health is recognized, articulated, and addressed. By bringing together Cyber Nursing, the Fundamentals of Care, and actor–network theory, the analysis has moved beyond viewing AI as a tool, instead conceptualizing it as a nonhuman actor that participates in the configuration of care.

A central argument has been that sexual health remains a form of weakly networked care—theoretically recognized yet inconsistently enacted. The introduction of AI does not resolve this marginalization but alters how it is expressed. Through care-like enactments, AI-mediated interactions may facilitate articulation and lower thresholds for disclosure, particularly in domains characterized by silence or discomfort. However, such interactions remain distinct from ethical care, as they lack responsibility, relational continuity, and professional accountability.

The concept of reassembling care networks has been advanced to capture this shift. AI participates in reconfiguring how concerns enter care processes, making certain needs more visible and discussable. Yet this reassembly is partial and ambivalent. While articulation may be expanded, integration into sustained, relational care is not guaranteed. The network is reconfigured but not stabilized.

The implications of this analysis are both conceptual and practical. Conceptually, the paper contributes to nursing theory by offering a sociomaterial understanding of care in which visibility, articulation, and responsibility are distributed across evolving care networks rather than located solely within individual actors. Practically, it suggests that nursing increasingly involves recognizing, interpreting, and integrating concerns that may first emerge through digitally mediated interactions. Rather than reducing the importance of nursing, AI makes the integrative and relational work of nursing more visible.

Ultimately, the significance of AI in cancer nursing lies not in its capacity to replace care, but in how it exposes the changing conditions under which fundamental care becomes possible. Sexual health, as a fundamental care need that remains weakly networked despite broad theoretical recognition, provides a critical lens through which these transformations become visible. The concept of reassembling care networks therefore complements the continuing development of the Fundamentals of Care framework by directing attention not only to what constitutes fundamental care, but also to the sociomaterial conditions through which fundamental care is recognized, articulated, and sustained in practice (Kitson 2026). In this sense, AI represents not simply a technological challenge for nursing, but an opportunity to further develop the conceptual foundations of fundamental care under increasingly mediated conditions.



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### Author Contributions

**Mats Christiansen:** conceptualization, methodology, formal analysis, writing – original draft, writing – review and editing. **Lena Wettergren:** writing – review and editing.


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